

MSc

Course title: SM6 Bayes Methods
Lecturer: Geoff Nicholls
Checked by: George Deligiannidis

Date of this version: 08/03/20

Do not turn this page until you are told that you may do so

1. A book club with n members wants to decide what book to read next. They have a shortlist of B books with labels $\mathcal{B} = \{1, \dots, B\}$. Let $\mathcal{P}_{\mathcal{B}}$ be the set of all permutations of the labels in $\mathcal{B}$. For $i = 1, \dots, n$ the i 'th reader gives a ranked list of the books $y_i = (y_{i,1}, \dots, y_{i,B})$, $y_i \in \mathcal{P}_{\mathcal{B}}$, ranking them from most to least interesting. The data are $y = (y_1, \dots, y_n)$.

In a Plackett-Luce model each book $b = 1, \dots, B$ has interest measure $\theta_b > 0$. Let $\theta = (\theta_1, \dots, \theta_B)$, $\theta \in R^B$. Let $Y_i \in \mathcal{P}_{\mathcal{B}}$ denote the random ranking from the i 'th reader. In the Plackett-Luce model, given $Y_{i,1} = y_{i,1}, \dots, Y_{i,a-1} = y_{i,a-1}$, the a 'th entry (ie, the next entry) is decided by choosing book b with probability proportional to θ_b from the books $\mathcal{B} \setminus \{y_{i,1}, \dots, y_{i,a-1}\}$ remaining. The $Y_1, \dots, Y_n$ are jointly independent given θ .

(a) [10 marks]

(i) Show that the likelihood $L(\theta; y)$ is

$$L(\theta; y) = \prod_{i=1}^n \prod_{a=1}^B \frac{\theta_{y_{i,a}}}{\sum_{b=a}^B \theta_{y_{i,b}}}.$$

Solution: (4 marks. Standard - this model appeared in the mock-assessed practical this year, with a different parameterisation)
Since

$$\begin{aligned} \Pr\{Y_i = y_i | \theta\} &= \Pr\{Y_{i,1} = y_{i,1} | \theta\} \times \Pr\{Y_{i,2} = y_{i,2} | Y_{i,1} = y_{i,1}, \theta\} \times \dots \\ &\quad \times \Pr\{Y_{i,B} = y_{i,B} | Y_{i,1:(B-1)} = y_{i,1:(B-1)}, \theta\} \end{aligned}$$

and

$$\Pr\{Y_{i,a} = y_{i,a} | Y_{i,1:(a-1)} = y_{i,1:(a-1)}, \theta\} = \frac{\theta_{y_{i,a}}}{\sum_{b=a}^B \theta_{y_{i,b}}}$$

from the definition of the process, we must have

$$\Pr\{Y_i = y_i | \theta\} = \prod_{a=1}^B \frac{\theta_{y_{i,a}}}{\sum_{b=a}^B \theta_{y_{i,b}}}, \quad i = 1, \dots, n$$

and therefore $L(\theta; y) = \prod_{i=1}^n \Pr\{Y_i = y_i | \theta\}$ by independence.

- (ii) The prior is $\pi_{\mathcal{B}}(\theta) = \prod_{b=1}^B \pi(\theta_b)$ with $\pi(\theta_b) = \Gamma(\theta_b; \alpha', 1)$ with $\alpha' > 0$ given. Write down the posterior density $\pi(\theta|y)$ and give an MCMC algorithm targeting $\pi(\theta|y)$.

Solution: (4 marks - bookwork)

First, $\pi(\theta|y) \propto \pi_{\mathcal{B}}(\theta)L(\theta; y)$ with $\pi_{\mathcal{B}}$ and L given.

Secondly, here is “an MCMC algorithm” targeting $\pi(\theta|y)$. Let $X_t = \theta$.

1. For $b = 1, \dots, B$

(a) simulate $\theta'_b \sim \pi(\cdot)$ and set $\theta'_a = \theta_a$, $a = 1, \dots, B$, $a \neq b$;

(b) with probability

$$\alpha(\theta'|\theta) = \min \left\{ 1, \frac{L(\theta'; y)}{L(\theta; y)} \right\}$$

set $\theta \leftarrow \theta'$.

2. Set $X_{t+1} = \theta$.

Some further simplification is possible but unnecessary.

- (iii) Explain why the scale β' in the prior $\Gamma(\alpha', \beta')$ for θ_i , $i \in \mathcal{B}$ may be set equal one. Suppose odds of 1000 : 1 for ranking one book above another represent extreme preference and are a priori unlikely for books on the shortlist. Explain how a fixed numerical value of α' might be chosen, noting any assumptions.

Solution: (2 marks - standard)

First, we can choose scale $\beta' = 1$. If we choose otherwise we can simply rescale $\theta' = \theta\beta'$. The likelihood is invariant so this factor is not identifiable anyway. Put another way, we can recover the posterior for $\beta' = c$ by multiplying $\theta \sim \pi(\theta|y, \beta' = 1)$ by c .

Secondly, suppose there are just $B = 2$ books in the list and one reader. I assume (very reasonably!) that prior knowledge about θ is not changed by these choices. Then

$$O_{1,2} = \Pr\{Y_1 = (1, 2)|\theta\} / \Pr\{Y_1 = (2, 1)|\theta\} = \theta_1/\theta_2.$$

If a value of $O_{1,2}$ as large as 1000 represents extreme preference across readers then it should be unlikely. I assume “unlikely” to mean having probability less than 1%. Choose α' so that the 99% quantile of θ_1/θ_2 is at 1000 (I assume this is possible - it is). This could be done using simulation if necessary. Many different choices and justifications may be given but concrete assumptions are required.

- (b) [10 marks] Suppose B is large so each reader $i = 1, \dots, n$ only reports the first N entries $x_i = (x_{i,1}, \dots, x_{i,N})$ in their ranking, with $N \ll B$. Here $x_{i,j} = y_{i,j}$ for $i = 1, \dots, n$ and $j = 1, \dots, N$. The data are $x = (x_1, \dots, x_n)$.

- (i) Show that the likelihood $L(\theta; x)$ for the new data is

$$L(\theta; x) = \prod_{i=1}^n \prod_{a=1}^N \frac{\theta_{x_{i,a}}}{\sum_{b=a}^N \theta_{x_{i,b}} + \sum_{d \in \mathcal{B} \setminus x_i} \theta_d}.$$

Solution: (3 marks - new)

From the process description we just take the first N outcomes but choose from the full list, so

$$\Pr\{Y'_{i,a} = x_{i,a} | Y'_{i,1:(a-1)} = x_{i,1:(a-1)}, \theta\} = \frac{\theta_{x_{i,a}}}{\sum_{b=a}^N \theta_{y_{i,b}} + \sum_{d \in \mathcal{B} \setminus y_i} \theta_d},$$

proceeding in the same way as before. Note that, for reader i , we always add $\sum_{d \in \mathcal{B} \setminus y_i} \theta_d$ in the denominator since these books didn't get picked by reader i so they never appear in the numerator. Taking the product,

$$\Pr\{Y'_i = x_i | \theta\} = \prod_{a=1}^N \frac{\theta_{y_{i,a}}}{\sum_{b=a}^N \theta_{y_{i,b}} + \sum_{d \in \mathcal{B} \setminus y_i} \theta_d}, \quad i = 1, \dots, n$$

with $L(\theta; x) = \prod_{i=1}^n \Pr\{Y'_i = x_i | \theta\}$.

- (ii) Let $\mathcal{C} = \bigcup_{i=1}^n x_i$ give the books appearing in at least one ranking and $\mathcal{D} = \mathcal{B} \setminus \mathcal{C}$ be the books appearing in none. Let $\theta_{\mathcal{C}} = (\theta_b)_{b \in \mathcal{C}}$ and $V = \sum_{d \in \mathcal{D}} \theta_d$. Write down the prior distribution of V and the likelihood $L(\theta_{\mathcal{C}}, V; x)$, and give the posterior $\pi(\theta_{\mathcal{C}}, V|x)$ as a function of $\theta_{\mathcal{C}}$ and V .

Solution: (4 marks - new)

First, $V \sim \Gamma(m\alpha', 1)$ with $m = |\mathcal{D}|$ in the prior.

Secondly, since $\mathcal{B} \setminus y_i = (\mathcal{C} \setminus y_i) \cup \mathcal{D}$ and the sets in the union are disjoint, the likelihood can be written

$$L(\theta_{\mathcal{C}}, V; x) = \prod_{i=1}^n \prod_{a=1}^N \frac{\theta_{y_i,a}}{\sum_{b=a}^N \theta_{y_i,b} + \sum_{d \in \mathcal{C} \setminus y_i} \theta_d + V}.$$

Only $\theta_b, b \in \mathcal{C}$ appear in this expression outside V so the parameterisation on the left is correct.

Thirdly, the posterior

$$\pi(\theta_{\mathcal{C}}, V|x) \propto L(\theta_{\mathcal{C}}, V; x) \Gamma(V; m\alpha', 1) \prod_{i \in \mathcal{C}} \pi(\theta_i),$$

has the same marginal distribution for $\theta_{\mathcal{C}}$ as $\pi(\theta|x)$.

Alternatively for this last bit, a more explicit calculation integrates $\pi(\theta|x)$ over dof in $\theta_{\mathcal{D}}|V$. These do not appear in the likelihood. We can write

$$\pi_{\mathcal{B}}(\theta) = \pi_{\mathcal{C}}(\theta_{\mathcal{C}}) \pi_{\mathcal{D}}(\theta_{\mathcal{D}})$$

and

$$\pi_{\mathcal{D}}(\theta_{\mathcal{D}}) = p(V) p(\theta_{\mathcal{D}}|V)$$

and use the fact that we know the marginal prior $p(V) = \Gamma(V; m\alpha', 1)$.

- (iii) Give an MCMC algorithm targeting $\pi(\theta_{\mathcal{C}}, V|x)$. State briefly why it may be more efficient, for estimation of $\theta_{\mathcal{C}}$ in the case $|\mathcal{C}| \ll B$, than MCMC targeting $\pi(\theta|x)$.

Solution: (3 marks - standard given above)

First the MCMC. Let $X_t = (\theta_{\mathcal{C}}, V)$.

1. For $b \in \mathcal{C}$

- (a) simulate $\theta'_b \sim \pi(\cdot)$ and set $\theta'_a = \theta_a$, $a \in \mathcal{C}$, $a \neq b$; this determines $\theta'_{\mathcal{C}}$
- (b) with probability

$$\alpha(\theta'_{\mathcal{C}}|\theta_{\mathcal{C}}) = \min \left\{ 1, \frac{L(\theta'_{\mathcal{C}}, V; x)}{L(\theta_{\mathcal{C}}, V; x)} \right\}$$

set $\theta_{\mathcal{C}} \leftarrow \theta'_{\mathcal{C}}$.

2. (a) simulate $V' \sim \Gamma(m\alpha', 1)$
- (b) with probability

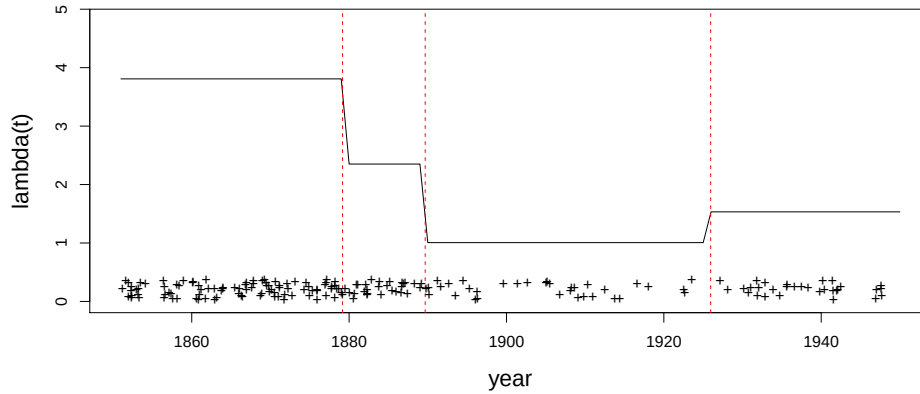
$$\alpha(V'|V) = \min \left\{ 1, \frac{L(\theta_{\mathcal{C}}, V'; x)}{L(\theta_{\mathcal{C}}, V; x)} \right\}$$

set $V' \leftarrow V$.

3. Set $X_{t+1} = (\theta_C, V)$.

Secondly, the efficiency remark. This target has a lower parameter dimension ($|\mathcal{C}| + 1 \ll B$) than $\pi(\theta|x)$ and similar computational cost per update so we expect more rapid MCMC convergence in a given computing time. Essentially we are targeting a marginal with uninteresting parameters integrated out.

2. During the 19th and 20th century there were many mining disasters. Let $L = 1850$ and $U = 1950$ and for $i = 1, 2, \dots, n$, let $y_i \in (L, U)$ be the date of the i 'th event. Let $y = (y_1, \dots, y_n)$.



Model the event times y as the arrival times of a Poisson process of piecewise constant rate $\lambda(t)$ per year. Let $\theta_0 = L$ and $\theta_m = U$ and for $i = 1, \dots, m-1$ let $\theta_i \in (L, U)$ be the sorted change point times at which $\lambda(t)$ jumps up or down. For $i = 1, \dots, m$ let $\lambda_i \geq 0$ give the disaster rate over the interval $(\theta_{i-1}, \theta_i]$. The rate function $\lambda(t) = \lambda(t; \theta, \lambda)$ for y is

$$\lambda(t) = \sum_{i=1}^m \lambda_i \mathbb{I}_{\theta_{i-1} < t \leq \theta_i} \quad L < t < U.$$

The data and a realisation of $\lambda(t)$ with $m = 4$ are shown in the Figure, with event dates y (+ signs), change points (θ vertical lines) and $\lambda(t)$ itself (piecewise constant function).

Let $\theta = (\theta_1, \dots, \theta_{m-1})$ and $\lambda = (\lambda_1, \dots, \lambda_m)$. Model the change-point times θ as the arrival times of a Poisson process of unknown rate ρ per year. The number of intervals m is unknown. Prior densities $\pi_R(\rho)$, $\rho \in [0, \infty)$ and $\pi_\Lambda(\lambda|m) = \prod_{i=1}^m \pi_\Lambda(\lambda_i)$, $\lambda \in [0, \infty)^m$ are given.

(a) [8 marks]

- (i) Give a rejection-ABC algorithm sampling $\pi_{ABC}(m, \theta, \lambda, \rho|y)$, the ABC approximation to the posterior $\pi(m, \theta, \lambda, \rho|y)$. Your answer should include summary-statistics and a distance measure. Assume π_R and π_Λ are easily simulated.

Solution: (6 marks. Bookwork - this example done in a practical)

First, a summary statistic. For example use histogram counts: take p a positive integer, let $c = (U - L)/p$, $b_0 = L$ and for $j = 1, \dots, p$ set $b_j = b_0 + cj$. Let $S(y) = (S_1(y), \dots, S_p(y))$ with $S_j(y) = \sum_{i=1}^N \mathbb{I}_{b_{j-1} < y_i \leq b_j}$. Here p should be chosen by experiment. Looking at the data the bandwidth is about 5 to 10 years so about $p = 10$ to 20 intervals over a century should yield a reasonably accurate ABC posterior. Here p can't be too large or many simulations K will be needed below.

Next, distance measure. Let $d(S, S') = \sum_{j=1}^p |S_j - S'_j|$.

Finally, rejection-ABC. Let $s = S(y)$ and $k = 1$. For $k = 1, \dots, K$:

1. simulate $\rho \sim \pi_R(\cdot)$, $m-1 \sim \text{Poisson}((U-L)\rho)$, $\theta_j \sim U(L, U)$ iid for $j = 1, \dots, m-1$ (and then sorted, with $\theta_0 = L$ and $\theta_m = U$) and $\lambda_i \sim \pi_\Lambda(\cdot)$ iid for $i = 1, \dots, m$;
2. for $i = 1, \dots, m$ simulate $n'_i \sim \text{Poisson}((\theta_i - \theta_{i-1})\lambda_i)$ and $y'_{i,j} \sim U(\theta_{i-1}, \theta_i)$ iid for $j = 1, \dots, n'_i$; set $y' = (y'_{1,1}, \dots, y'_{m,n'_m})$;

3. compute $s' = S(y')$ and $d = d(s, s')$;

4. store the simulated values $\rho_k = \rho$, $m_k = m$, $\tau_k = \tau$, $\lambda_k = \lambda$ and $d_k = d$.

Let $\delta > 0$ be given. The rejection ABC sample is $X = \{(\theta_k, \lambda_k, m_k, \rho_k) : d_k < \delta, k = 1, \dots, K\}$. An answer choosing δ as a quantile of $d_1, \dots, d_K$ is also accepted.

- (ii) It is of interest to estimate the unknown true rate $\lambda(t)$ as a function of $t \in (L, U)$. State how you would use the ABC-sample output to estimate this function.

Solution: (2 marks. Bookwork - this example done in a practical)
Suppose $|X| = A$. For $a = 1, \dots, A$ let $X_a = (\theta^{(a)}, \lambda^{(a)}, m^{(a)}, \rho^{(a)})$ be the a 'th element of X taken in arbitrary fixed order. Use the ABC-posterior mean (the estimator minimising the (ABC-posterior) Bayes risk for the L2 loss at each t): for any $t \in (L, U)$,

$$\hat{\lambda}(t) = \frac{1}{A} \sum_{a=1}^A \lambda(t; \theta^{(a)}, \lambda^{(a)}).$$

(b) [6 marks]

- (i) Write down the prior $\pi(\theta, \lambda, m, \rho)$ in as much detail as you can. Specify its parameter space, $(\theta, \lambda, m, \rho) \in \Omega$ say.

Solution: (4 marks. New though all written down in practical handout as optional background.)

Let $T = U - L$. We have $\pi(m|\rho) = \text{Poisson}(m; T\rho)$ and for $\theta \in \Theta_{m-1}$,

$$\pi_{\Theta}(\theta|m) = (m-1)!/T^{m-1}$$

(recall they are ordered), with $\Theta_0 = \emptyset$ (when $m = 1$) and

$$\Theta_{m-1} = \{\theta \in [0, \infty)^{m-1} : L < \theta_1 < \dots < \theta_{m-1} < U\}.$$

The full prior is

$$\begin{aligned} \pi(\theta, \lambda, m, \rho) &= \pi_{\Theta}(\theta|m)\pi_{\Lambda}(\lambda|m)\pi(m|\rho)\pi(\rho) \\ &\propto \frac{(m-1)!}{T^{m-1}}\pi_{\Lambda}(\lambda|m)\exp(-T\rho)\frac{(T\rho)^{m-1}}{(m-1)!}\pi(\rho) \\ &\propto \exp(-T\rho)\rho^{m-1}\pi_{\Lambda}(\lambda|m)\pi(\rho) \end{aligned}$$

with $\pi_{\Lambda}(\lambda|m)\pi(\rho)$ given in the question. The parameter space for $(\theta, \lambda, m, \rho)$ is

$$\Omega = [0, \infty) \times \bigcup_{m=1}^{\infty} \{m\} \times \Lambda_m \times \Theta_{m-1}$$

where $\Lambda_m = [0, \infty)^m$.

- (ii) Write down the posterior $\pi(\lambda, \theta, m, \rho|y)$ in terms of the available model elements.

Solution: (2 marks. Standard) Let $n_j = \sum_i \mathbb{I}_{\theta_{j-1} < y_i \leq \theta_j}$ be the number of events in the interval $(\theta_{j-1}, \theta_j]$. This number is $\text{Poisson}(\lambda_j(\theta_j - \theta_{j-1}))$ and the points are otherwise scattered uniformly at random. The likelihood is

$$\begin{aligned} L(\theta, \lambda; y) &= \prod_{j=1}^m e^{-\lambda_j(\theta_j - \theta_{j-1})} \frac{(\lambda_j(\theta_j - \theta_{j-1}))^{n_j}}{n_j!} \times \frac{1}{(\theta_j - \theta_{j-1})^{n_j}} \\ &\propto \prod_{j=1}^m e^{-\lambda_j(\theta_j - \theta_{j-1})} \lambda_j^{n_j} \end{aligned}$$

(which alternatively follows from exponential intervals) so the posterior is

$$\pi(\lambda, \theta, m, \rho | y) \propto \pi(\theta, \lambda, m, \rho) L(\theta, \lambda; y)$$

with the likelihood and prior given above and parameter space $(\lambda, \theta, m, \rho) \in \Omega$ with Ω defined above.

- (c) [6 marks] In a reversible jump MCMC algorithm targeting $\pi(\lambda, \theta, m, \rho | y)$, birth and death updates are chosen with probabilities $p_{m,m+1}$ and $p_{m,m-1}$ respectively. A birth proposal $(\lambda, \theta, m, \rho) \rightarrow (\lambda', \theta', m', \rho)$ with $m' = m + 1$ is generated as follows: choose an interval $i \sim U\{1, \dots, m\}$ uniformly; simulate a split point $\theta^* \sim U(\theta_{i-1}, \theta_i)$; simulate two new values $\lambda_{i,1}, \lambda_{i,2} \sim \text{Exp}(1)$ independently. In the candidate state

$$\lambda' = (\lambda_1, \dots, \lambda_{i-1}, \lambda_{i,1}, \lambda_{i,2}, \lambda_{i+1}, \dots, \lambda_m)$$

and

$$\theta' = (\theta_1, \dots, \theta_{i-1}, \theta^*, \theta_i, \dots, \theta_{m-1}).$$

Give a matching death proposal $(\lambda', \theta', m', \rho) \rightarrow (\lambda, \theta, m, \rho)$ and the acceptance probability for the birth proposal. No simplification of expressions is required.

Solution: (6 marks. New - we did reversible jump for a normal mixture model, and the death update here doesn't have an atom on the proposed state).

Relabel $\theta' = (\theta'_1, \dots, \theta'_{m'-1})$ and $\lambda' = (\lambda'_1, \dots, \lambda'_{m'})$ and consider now generating $(\lambda, \theta, m, \rho)$ as a candidate. The death update is chosen with probability $p_{m',m}$. In the death update choose $i \sim U\{1, \dots, m' - 1\}$, delete θ'_i and λ'_i and λ'_{i+1} . Simulate a new value $\tilde{\lambda}_i \sim \text{Exp}(1)$ say. Set $\lambda = (\lambda'_1, \dots, \lambda'_{i-1}, \tilde{\lambda}_i, \lambda'_{i+2}, \dots, \lambda'_{m'})$ and $\theta = (\theta'_1, \dots, \theta'_{i-1}, \theta'_{i+1}, \dots, \theta'_{m'})$.

The probability distribution for the birth proposal is

$$q(\lambda', \theta', m', \rho | \lambda, \theta, m, \rho) = p_{m,m+1} m^{-1} (\theta_i - \theta_{i-1})^{-1} \text{Exp}(\lambda_{i,1}; 1) \text{Exp}(\lambda_{i,2}; 1)$$

and for the death proposal is

$$q(\lambda, \theta, m, \rho | \lambda', \theta', m', \rho) = p_{m+1,m} m^{-1} \text{Exp}(\lambda_i; 1).$$

Notice that the death proposal has to hit the old state, so the proposal density for $\tilde{\lambda}_i$ is evaluated at $\tilde{\lambda}_i = \lambda_i$.

The acceptance probability for the birth is simply

$$\alpha(\lambda', \theta', m', \rho | \lambda, \theta, m, \rho) = \min \left\{ 1, \frac{\pi(\lambda', \theta', m', \rho | y) q(\lambda, \theta, m, \rho | \lambda', \theta', m', \rho)}{\pi(\lambda, \theta, m, \rho | y) q(\lambda', \theta', m', \rho | \lambda, \theta, m, \rho)} \right\}.$$

The following remarks are not needed if the above is correct, though credit may be awarded if the above is in error but some consideration is given to dimension-matching in the answer. The dimensions balance since

$$\dim(\theta, \lambda) + \dim(\theta^*, \lambda_{i,1}, \lambda_{i,2}) = m - 1 + m + 3 = 2m + 2$$

and

$$\dim(\theta', \lambda') + \dim(\tilde{\lambda}_i) = m' - 1 + m' + 1 = 2m' = 2m + 2.$$

The Jacobian

$$\left| \frac{\partial(\theta', \lambda', \tilde{\lambda}_i)}{\partial(\theta, \lambda, \theta^*, \lambda_{i,1}, \lambda_{i,2})} \right| = 1$$

because all the elements in $(\theta', \lambda', \tilde{\lambda}_i)$ are matched to, and equal to, unique elements in $(\theta, \lambda, \theta^*, \lambda_{i,1}, \lambda_{i,2})$. Adding and deleting entries in a vector renumbers some entries in the new vector. Our notation avoids making this explicit.